Co-ordination and response

20 questions: 7 multiple choice, 13 structured.

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1 · Part 6

Paper 1 & 2 | Co-ordination and response | [1]

Which sequence shows the correct order of events in a co-ordinated response?

A receptor → stimulus → effector → response

B stimulus → receptor → effector → response

C stimulus → effector → receptor → response

D effector → stimulus → receptor → response

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2 · Part 6

Paper 1 & 2 | Co-ordination and response | [1]

What does the central nervous system consist of?

A the brain and the sense organs

B the brain and the nerves

C the spinal cord and the nerves

D the brain and the spinal cord

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3 · Part 6

Paper 1 & 2 Co-ordination and response [1]

The four events listed below happen when an electrical impulse passes from one neurone to the next neurone at a synapse.

1 the neurotransmitter binds to receptors on the next neurone

2 an impulse arrives at the end of the first neurone

3 the neurotransmitter diffuses across the gap

4 the neurotransmitter is released

Which is the correct order of these events?

A 2 → 4 → 3 → 1

B 4 → 2 → 3 → 1

C 2 → 3 → 4 → 1

D 2 → 4 → 1 → 3

4 · Part 6

Paper 1 & 2 Co-ordination and response [1]

Ravi is reading a book. He looks up to watch a bird in a tree far away.

Which row shows the changes in his eyes as he focuses on the bird?

	ciliary muscles	suspensory ligaments	lens
A	contract	slacken	becomes fatter
B	contract	tighten	becomes thinner
C	relax	tighten	becomes thinner
D	relax	slacken	becomes fatter

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5 · Part 6

Paper 2 only | Co-ordination and response | [1]

Which row shows the source of ADH and the source of oestrogen?

	source of ADH	source of oestrogen
A	pituitary gland	testis
B	pituitary gland	ovary
C	kidney	ovary
D	kidney	uterus

6 · Part 6

Paper 1 & 2 | Co-ordination and response | [1]

Three young oat seedlings, X, Y and Z, were each given light from one side only for two days.

In seedling X, the tip of the shoot was cut off.

In seedling Y, the tip of the shoot was covered with a small cap that does not let light through.

In seedling Z, the shoot was not treated.

Which seedlings bent towards the light?

A X and Y only

B Y and Z only

C X only

D Z only

7 · Part 6

Paper 2 only | Co-ordination and response | [1]

Which hormone stimulates the release of an egg from an ovary?

A LH

B FSH

C oestrogen

D progesterone

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8 · Part 6

Paper 1 & 2 | Co-ordination and response | [7]

Organisms are able to respond to changes in their environment. Some changes inside the body are also controlled.

(a) State what is meant by the term homeostasis. [1]

(b) Give two examples of homeostasis in the human body.

Woodlice are small animals that lose water easily through their body surface. They usually live in damp places under logs and stones.

When a woodlouse is in dry air, it moves around faster. When it reaches damp air, it slows down and stays still. [2]

(c) Identify the stimulus and the effector in this response. [2]

(d) Explain how this response helps the woodlouse to survive. [2]

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9 · Part 6

Paper 1 & 2 | Co-ordination and response | [6]

Leila germinated some bean seeds in a dark cupboard. When each seedling had a short root and a short shoot, she laid it on its side on damp cotton wool, so that both the root and the shoot were horizontal. She then kept the seedlings in the dark cupboard for two more days.

(a) Describe how the root and the shoot would grow during the two days. [2]

(b) State the name of this type of plant response. [1]

(c) Explain why Leila kept the seedlings in the dark during this investigation.

Leila then grew some other seedlings in a clear container of water, so that their roots were exposed to light. Light reached the container from one side only. [2]

(d) Describe the phototropic response of the roots. [1]

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10 · Part 6

Paper 1 & 2 | Co-ordination and response | [7]

In an investigation into auxin, a scientist cut the tips off some young grass shoots. She stood the tips on small blocks of agar jelly for two hours, so that auxin diffused from the tips into the agar. Different blocks had different numbers of tips placed on them.

Each agar block was then placed on the left-hand side of the cut top of a different grass shoot. This shoot had already had its own tip removed. The shoots were kept in the dark for 24 hours and then the angle of bending of each shoot was measured.

The table shows the results.

number of tips placed on the agar block	angle of bending / degrees
0	0
1	7
2	13
3	17
4	18

(a) Describe the relationship between the number of tips placed on the agar block and the angle of bending. [2]

(b) Give a reason why each shoot had its own tip removed before the agar block was placed on it. [1]

(c) Give a reason why one agar block had no tips placed on it. [1]

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(d) Explain how auxin causes a shoot that is lit from one side to grow towards the light. [3]

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11 · Part 6 Paper 1 & 2 Co-ordination and response [6]

Responses in the human body are controlled by the nervous system and by hormones.

(a) Complete the table to compare nervous communication with hormonal communication.

feature	nervous communication	hormonal communication
type of signal	electrical impulses	.....
how the signal is carried	along neurones	.....
speed of the response	rapid	.....
how long the effect lasts	.....	long-lasting

[4]

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(b) Hormones are released by glands. Explain why one hormone can affect organs in several different parts of the body. [2]

12 · Part 6

Paper 1 & 2 | Co-ordination and response | [6]

Tom accidentally touches a hot pan. His finger moves away from the pan before he is aware of the heat. This is a reflex response.

(a) Describe the structure and functioning of the reflex arc that causes Tom's finger to move away from the hot pan. [4]

(b) Explain how this reflex response is an advantage to Tom. [2]

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13 · Part 6

Paper 1 & 2 | Co-ordination and response | [8]

Jasmine investigated whether listening to music affects reaction time.

Her partner held a 30 cm ruler vertically, with the zero mark level with the top of Jasmine's thumb. Without warning, her partner let go of the ruler and Jasmine caught it as quickly as she could. The distance the ruler fell was converted into a reaction time using a conversion table.

Jasmine did five trials in silence and five trials while listening to loud music through headphones.

The table shows her results.

trial	**1**	**2**	**3**	**4**	**5**
reaction time in silence / s	0.21	0.19	0.20	0.22	0.18
reaction time with music / s	0.25	0.24	0.38	0.26	0.25

(a) Identify the independent variable and the dependent variable in this investigation. [2]

(b) State two variables that Jasmine should keep the same. [2]

(c) Calculate the mean reaction time with music. Do not include the anomalous result.

Use the formula: mean = total of the results ÷ number of results [2]

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(d) Catching the ruler is not a reflex response. The light reaching Jasmine's eyes is the stimulus.

Explain why Jasmine's reaction time in this test is longer than the time taken for a reflex response. [2]

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14 · Part 6

Paper 1 & 2 Co-ordination and response [6]

The eye is a sense organ.

(a) Complete the table to give the function of each part of the eye.

part of the eye	function
cornea	.....
iris	.....
retina	.....
optic nerve	.....

[4]

(b) Explain why the eye is described as a receptor. [2]

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15 · Part 6

Paper 1 & 2 | Co-ordination and response | [6]

Sara measured the diameter of the pupil of a volunteer's eye at different light intensities.

The table shows her results.

light intensity / lux	pupil diameter / mm
10	7.0
100	6.0
500	4.6
2000	3.3
10 000	2.4

(a) Describe the relationship between light intensity and pupil diameter. [1]

(b) Calculate how many times greater the pupil diameter is at 10 lux than at 10 000 lux. [2]

(c) Explain how muscles in the eye cause the change in pupil diameter when light intensity increases. [2]

(d) Suggest why this response is important in very bright light. [1]

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16 · Part 6

Paper 1 & 2 | Co-ordination and response | [7]

Daniel cycled on an exercise bike in a warm room for 30 minutes. His core body temperature and his rate of sweat production were measured every 10 minutes.

The table shows the results.

time / min	core body temperature / °C	rate of sweat production / g per min
0	37.0	0.2
10	37.4	1.4
20	37.6	2.9
30	37.6	3.1

(a) Describe how Daniel's rate of sweat production changed during the 30 minutes. [2]

(b) Explain how sweating helps to stop Daniel's core body temperature rising further. [2]

(c) During the exercise, the skin on Daniel's face and arms looked redder than before.

Explain how changes in the blood vessels in his skin help Daniel to lose heat. [3]

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17 · Part 6

Paper 1 & 2 | Co-ordination and response | [6]

Kofi is about to run a race. He feels nervous, and the hormone adrenaline is released into his blood.

Kofi's heart rate was measured while he was resting on the day before the race, and again just before the race started.

time of measurement	heart rate / beats per minute
resting, day before the race	64
just before the race	88

(a) Name the glands that release adrenaline. [1]

(b) Calculate the percentage increase in Kofi's heart rate.

Use the formula: percentage increase = (new value − original value) ÷ original value × 100 [2]

(c) Explain how the increase in heart rate helps the muscles in Kofi's legs during the race. [2]

(d) Give one other effect of adrenaline on the body. [1]

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18 · Part 6

Paper 1 & 2 Co-ordination and response [7]

Two people each drank a glucose drink at time 0. Their blood glucose concentration was then measured every 30 minutes.

Person P produces normal amounts of insulin. Person Q produces very little insulin.

time / min	blood glucose concentration of P / mmol per dm^3	blood glucose concentration of Q / mmol per dm^3
0	5.0	7.0
30	7.8	12.4
60	6.6	14.8
90	5.4	14.3
120	5.0	13.4
150	4.9	12.6

(a) State the source of insulin in the body. [1]

(b) Calculate the rate of decrease in the blood glucose concentration of person P between 30 minutes and 90 minutes.

Use the formula: rate = change in concentration ÷ time taken [2]

(c) Explain the changes in the blood glucose concentration of person P between 30 minutes and 120 minutes. [3]

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(d) Suggest why the blood glucose concentration of person Q stays high. [1]

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19 · Part 6

Paper 2 only

Co-ordination and response

[8]

Mei's urine was tested twice on the same day.

In the morning she had been drinking normally. In the afternoon she had been working outdoors on a hot day. She had sweated a lot and had not drunk anything for several hours.

	morning	**afternoon**
volume of urine produced per hour / cm^3	75	20
concentration of urine / arbitrary units	280	1050

(a) Describe how Mei's urine in the afternoon was different from her urine in the morning. [2]

(b) Explain how the hormone ADH caused the changes in Mei's urine in the afternoon. [4]

(c) In the evening, Mei drank a large volume of water.

Predict how the amount of ADH released into her blood would change. Give a reason for your answer. [2]

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20 · Part 6

Paper 2 only

Co-ordination and response

[7]

The table shows the concentrations of three hormones in the blood of a woman during one menstrual cycle. Day 1 is the first day of menstruation (bleeding).

day of cycle	FSH / arbitrary units	oestrogen / arbitrary units	progesterone / arbitrary units
1	4	2	1
4	5	3	1
8	4	7	1
12	6	14	1
14	9	9	2
16	3	5	5
20	2	7	12
24	2	4	7
28	4	2	2

(a) State two roles of FSH in the menstrual cycle. [2]

(b) Name the gland that releases FSH. [1]

(c) Describe how the concentration of progesterone changes between day 14 and day 28. [2]

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(d) Explain the role of progesterone in the second half of the menstrual cycle, and what happens when its concentration falls. [2]

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